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United States Patent Application Publication

20250286622

Kind Code

A1

Publication Date

September 11, 2025

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OPTICAL FIBER DISTRIBUTED WIRELESS SIGNAL COVERAGE SYSTEM BASED ON ROF TECHNOLOGY

Abstract

Disclosed is an optical fiber distributed wireless signal coverage system based on a radio over fiber (ROF) technology, including a signal access intelligent unit, configured to perform radio frequency signal magnitude conditioning and then radio frequency amplification on a first wireless signal to obtain a first downlink electrical signal and combine the first downlink electrical signal, an OOK modulating signal, and a first radio frequency signal together and convert the same into a downlink laser signal, and output the signal to a first optical fiber interface; and further configured to convert a resultant uplink laser signal into an uplink electrical signal and perform radio frequency amplification and radio frequency signal magnitude conditioning on the uplink electrical signal to output the signal from a radio frequency interface; an optical fiber splitter, configured to divide the downlink laser signal into downlink laser sub-signals; and at least one multi-standard remote radio unit.

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Family ID: 1000008505408

Appl. No.: 19/063935

Filed: February 26, 2025

Foreign Application Priority Data

CN

202410268404.2

Mar. 10, 2024

Related U.S. Application Data

Publication Classification

Int. Cl.: H04B10/2575 (20130101); H04B10/50 (20130101); H04L27/04 (20060101)

U.S. Cl.:

CPC H04B10/25759 (20130101); H04B10/503 (20130101); H04L27/04 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure belongs to the technical field of Radio Over Fiber (ROF) and wireless signal coverage systems and mainly relates to an ROF optical transceiver part, an optical passive part, a wireless system part, a software control part, and a software monitoring part, specifically an optical fiber distributed wireless signal coverage system based on an ROF technology.

BACKGROUND

[0002] At present, a wireless mobile communication technology increasingly affects the modern life of people. From receiving and transmitting WeChat messages, watching short videos, microblogging, issuing TikTok videos, network payments, going out with map navigation, to watching high-definition televisions and videos at home, a mobile network of a smart phone makes people enjoy colorful life. With the increase of mobile data traffic and increasing tools and devices working in a plurality of communication platforms through channels of high bandwidth mobile communication, demand modes on Internet change. People also hope that wireless network communication supports more users and covers a wider area while hoping to enjoy high bandwidth of wireless communication, and really hope signal coverage without a signal dead angle to realize ubiquitous high-speed wireless signal coverage indeed. The increase of signal bandwidth and ever increase in the number of device users bring a huge challenge to design and architecture of a wireless signal coverage system and the operation of a future broadband wireless network. The user access bandwidth is improved and more user accesses are supported by reducing a signal coverage range of the single mobile network, constructing a small coverage area, and improving the carrier frequency and signal intensity of wireless communication.

SUMMARY

[0003] To overcome deficiencies in the prior art, the present disclosure provides an optical fiber distributed wireless signal coverage system based on a ROF technology with low cost, high speed, and low power consumption.

[0004] The technical solution of the present disclosure is as follows:

[0005] An optical fiber distributed wireless signal coverage system based on a ROF technology is provided, including: [0006] a signal access intelligent unit, including a radio frequency interface and a first optical fiber interface and configured to perform radio frequency signal magnitude conditioning and then radio frequency amplification on a first wireless signal received from the radio frequency interface to obtain a first downlink electrical signal and combine the first downlink electrical signal, an OOK modulating signal, and a first radio frequency signal together and convert the same into a downlink laser signal, and output the downlink laser signal to the first optical fiber interface; and further configured to convert a resultant uplink laser signal inputted from the first optical fiber interface into an uplink electrical signal and perform radio frequency amplification

and radio frequency signal magnitude conditioning on the uplink electrical signal in sequence to output the uplink electrical signal from the radio frequency interface; [0007] an optical fiber splitter, including a first optical port and at least one second optical port, where the first optical port thereof is connected to the first optical fiber interface through a first optical fiber; configured to averagely divide the downlink laser signal inputted through the first optical port into a plurality of downlink laser sub-signals, where the downlink laser sub-signals correspond to the second optical ports one by one and are outputted from the second optical ports; and further configured to combine the uplink laser signals inputted through the second optical ports into a resultant uplink laser signal and output the resultant uplink laser signal from the first optical port; and [0008] at least one multi-standard remote radio unit, the multi-standard remote radio units including second optical fiber interfaces respectively, where the second optical fiber interfaces are respectively connected to the second optical ports of the optical fiber splitter in a one-to-one correspondence manner through a second optical fiber; configured to decompose the downlink laser sub-signals and convert the downlink laser sub-signals into second downlink electrical signals and divide the second downlink electrical signals into a first signal, a second signal, and a third signal, and perform radio frequency signal magnitude conditioning and radio frequency amplification on the third signal in sequence and transmit the third signal; and further configured to filter a received second wireless signal to obtain an uplink radio frequency signal, perform low-noise amplification and radio frequency signal magnitude conditioning on the uplink radio frequency signal in sequence and combine the uplink radio frequency signal with a second radio frequency signal and convert the same into an uplink laser signal, and transmit the uplink laser signal to the second optical fiber interfaces.

[0009] The signal access intelligent unit further includes: [0010] a first radio frequency switch/filter, input and output ports thereof being electrically connected to a second port of the radio frequency interface; [0011] a first downlink ATT, an input port thereof being electrically connected to the output port of the first radio frequency switch/filter; [0012] a downlink amplifier, an input port thereof being electrically connected to an output port of the first downlink ATT; [0013] a first signal coupler, a first input port thereof being electrically connected to an output port of the downlink amplifier; [0014] a large dynamic ROF optical transmitter, an input port thereof being electrically connected to an output port of the first signal coupler; [0015] an OOK modulator, an output port thereof being electrically connected to a second input port of the first signal coupler; [0016] a wavelength division multiplexer, an input port thereof being connected to an output port of the large dynamic ROF optical transmitter through a fourth optical fiber and input and output ports thereof being connected to the first optical fiber interface through a fifth optical fiber; [0017] a first ROF optical receiver, an input port thereof being connected to an output port of the wavelength division multiplexer through a sixth optical fiber; [0018] a second signal coupler, an input port thereof being electrically connected to an output port of the first ROF optical receiver; [0019] a first uplink amplifier, an input port thereof being electrically connected to a first output port of the second signal coupler; [0020] a first uplink ATT, an input port thereof being electrically connected to an output port of the first uplink amplifier and an output port thereof being electrically connected to the input port of the first radio frequency switch/filter; [0021] a first communication module, an input port thereof being electrically connected to a second output port of the second signal coupler and an output port thereof being electrically connected to a third input port of the first signal coupler; and [0022] a master monitor unit, a data port thereof being connected to a data port of the first radio frequency switch/filter, a data port of the first downlink ATT, a data port of the downlink amplifier, a data port of the large dynamic ROF optical transmitter, a data port of the OOK modulator, a data port of the first communication module, a data port of the first uplink ATT, a data port of the first uplink amplifier, and a data port of the first ROF optical receiver through a first data line.

[0023] The optical fiber splitter further includes: [0024] a $1 \times N$ optical splitter, an input optical port

thereof serving as the first optical port of the optical fiber splitter; and [0025] at least one 1×8 optical splitter, input optical ports thereof being connected to output optical ports of the $1 \times N$ optical splitter in a one-to-one correspondence manner through the third optical fiber and output optical ports thereof serving as the second optical ports of the optical fiber splitter.

[0026] The multi-standard remote radio unit further includes: [0027] a wavelength division demultiplexer, input and output ports thereof being connected to the second optical fiber interface through a tenth optical fiber; [0028] a second ROF optical receiver, an input port thereof being connected to an output port of the wavelength division demultiplexer through an eighth optical fiber; [0029] a third signal coupler, an input port thereof being electrically connected to an output port of the second ROF optical receiver; [0030] a second downlink ATT, an input port thereof being electrically connected to a first output port of the third signal coupler; [0031] a downlink PA, an input port thereof being electrically connected to an output port of the second downlink ATT; [0032] a second radio frequency switch/filter, an input port thereof being electrically connected to an output port of the downlink PA; [0033] a filter, a first port thereof being electrically connected to input and output ports of the second radio frequency switch/filter; [0034] an antenna, electrically connected to a second port of the filter; [0035] an uplink LNA, an input port thereof being electrically connected to the output port of the second radio frequency switch/filter; [0036] a second uplink ATT, an input port thereof being electrically connected to an output port of the uplink LNA; [0037] a fourth signal coupler, a first input port thereof being electrically connected to an output port of the second uplink ATT; [0038] a variable wavelength ROF optical transmitter, an input port thereof being electrically connected to an output port of the fourth signal coupler and an output port thereof being electrically connected to the input port of the wavelength division demultiplexer through a ninth optical fiber; [0039] an OOK demodulator, an input port thereof being electrically connected to a second output port of the third signal coupler and a data interface thereof being connected to the second downlink ATT and the downlink PA through a third data line, respectively; [0040] a second communication module, an input port thereof being electrically connected to a third output port of the third signal coupler; and [0041] a slave monitor unit, a data port thereof being connected to a data port of the second ROF optical receiver, a data port of the second downlink ATT, a data port of the downlink PA, a data port of the OOK demodulator, a data port of the second communication module, a data port of the uplink LNA, a data port of the second uplink ATT, and a data port of the variable wavelength ROF optical transmitter through a second data line.

[0042] The present disclosure has the beneficial effects as follows:

[0043] The system provided by the present disclosure is simple in architecture, convenient in engineering construction and engineering maintenance, and the wireless coverage system can be rapidly deployed by using an existing optical fiber network at present. The system provided by the present disclosure cannot only transmit WIFI, 4G, and 5G signals, but also transmit millimeter wave 5G signals and future 6G signals, and can perform smooth transition, thereby facilitating the update of communication systems. By adopting the large dynamic ROF optical transmission technology, the present disclosure solves a problem of optical insertion loss caused by the optical splitter and long-distance transmission. By adopting the large dynamic variable wavelength ROF optical transmission technology, the present disclosure solves a problem of laser interference of lasers of the plurality of multi-standard remote radio units **3** in an uplink optical combined path. The optical fiber distributed wireless signal coverage system based on an ROF technology provided by the present disclosure includes a signal access intelligent unit **1**, an optical fiber splitter **2**, and a multi-standard remote radio unit **3**. The three devices are connected only with a common single model fiber, so that the cost is low and it is convenient to wire. Optical fiber connection between the signal access intelligent unit **1** and the optical fiber splitter **2** may adopt long-distance transmission of an optical fiber cable according to an actual wireless signal coverage demand. The present disclosure may support the highest 20 km transmission. Optical fiber connection between

the optical fiber splitter **2** and the multi-standard remote radio unit **3** can be connection with a 200-300 m optical fiber usually according to an actual wireless signal coverage demand for signal coverage in each area indoors.

Description

BRIEF DESCRIPTION OF FIGURES

[0044] FIG. **1** is a system architecture diagram of the present disclosure;

[0045] FIG. **2** is a functional block diagram of a signal access intelligent unit of the present disclosure;

[0046] FIG. **3** is a functional block diagram of a multi-standard remote radio unit of the present disclosure;

[0047] FIG. **4** is a software control flowchart where a wavelength of the multi-standard remote radio unit **3** is set successfully;

[0048] FIG. **5** is a software control flowchart where a wavelength of the multi-standard remote radio unit **3** fails to set;

[0049] FIG. **6** is a system architecture diagram of a conventional digital wireless signal coverage solution;

[0050] FIG. **7** is a curve graph where third-order intercept point data is actually measured through a direct connection between a large dynamic ROF optical transmitter and an ROF optical receiver with 1 m optical fiber in the present disclosure;

[0051] FIG. **8** is a curve graph where gain data is actually measured through the direct connection between a large dynamic ROF optical transmitter and an ROF optical receiver with 1 m optical fiber in the present disclosure;

[0052] FIG. **9** is a curve graph where noise figure data is actually measured through the direct connection between a large dynamic ROF optical transmitter and an ROF optical receiver with 1 m optical fiber in the present disclosure; and

[0053] FIG. **10** is a curve graph where spurious-free dynamic range data is actually measured through a direct connection between a large dynamic ROF optical transmitter and an ROF optical receiver with 1 m optical fiber in the present disclosure.

[0054] Numerals in the drawings: **1**, signal access intelligent unit; **2**, optical fiber splitter; **3**, multi-standard remote radio unit; **4**, 1×N optical splitter; **5**, 1×8 optical splitter; **8**, first radio frequency switch/filter; **9**, first downlink ATT; **10**, downlink amplifier; **11**, first signal coupler; **12**, large dynamic ROF optical transmitter; **13**, OOK modulator; **14**, wavelength division multiplexer; **15**, first power supply module; **16**, first communication module; **17**, master monitor unit; **18**, first uplink ATT; **19**, first uplink amplifier; **20**, second signal coupler; **21**, first ROF optical receiver; **22**, wavelength division demultiplexer; **23**, second ROF optical receiver; **24**, third signal coupler; **25**, second downlink ATT; **26**, downlink PA; **27**, OOK demodulator; **28**, slave monitor unit; **29**, second communication module; **30**, second power supply module; **31**, second radio frequency switch/filter; **32**, filter; **33**, antenna; **34**, uplink LNA; **35**, second uplink ATT; **36**, fourth signal coupler; **37**, variable wavelength ROF optical transmitter; **38**, radio frequency interface; **39**, first optical fiber interface; **40**, second optical fiber interface; **42**, first optical port; and **43**, second optical port.

DETAILED DESCRIPTION

[0055] The present disclosure is further described below through specific embodiments, and is not limited thereto.

[0056] Referring to FIG. **1**, FIG. **2**, and FIG. **3**, an optical fiber distributed wireless signal coverage system based on an ROF technology in many embodiments of the present disclosure includes a signal access intelligent unit **1**, an optical fiber splitter **2**, and a multi-standard remote radio unit **3**.

[0057] The signal access intelligent unit **1** includes a radio frequency interface **38** and a first optical

fiber interface **39**. The signal access intelligent unit **1** is configured to perform radio frequency signal magnitude conditioning and then radio frequency amplification on a first wireless signal received from the radio frequency interface **38** to obtain a first downlink electrical signal and combine the first downlink electrical signal, an OOK modulating signal **F1**, and a first radio frequency signal **F2** together and convert the same into a downlink laser signal, and output the downlink laser signal to the first optical fiber interface **39**. The signal access intelligent unit **1** is further configured to convert a resultant uplink laser signal inputted from the first optical fiber interface **39** into an uplink electrical signal and perform radio frequency amplification and radio frequency signal magnitude conditioning on the uplink electrical signal in sequence to output the uplink electrical signal from the radio frequency interface **38**.

[0058] The optical fiber splitter **2** includes a first optical port **42** and at least one second optical port **43**, where the first optical port **42** of the optical fiber splitter **2** is connected to the first optical fiber interface **39** through a first optical fiber. The optical fiber splitter **2** is configured to averagely divide the downlink laser signal inputted through the first optical port **42** into a plurality of downlink laser sub-signals, where the downlink laser sub-signals correspond to the second optical ports **43** one by one and are outputted from the second optical ports **43**. The optical fiber splitter **2** is further configured to combine the uplink laser signals inputted through the second optical ports **43** into a resultant uplink laser signal and output the resultant uplink laser signal from the first optical port **42**.

[0059] In the embodiment shown in FIG. **1**, there are **M** multi-standard remote radio units **3**, **M** is an integer greater than or equal to 1, and the specific number is determined according to an actual demand. The multi-standard remote radio units **3** include second optical fiber interfaces **40**, where the second optical fiber interfaces **40** are respectively connected to the second optical ports **43** of the optical fiber splitter **2** in a one-to-one correspondence manner through a second optical fiber. The multi-standard remote radio units **3** are configured to decompose the downlink laser sub-signals and convert the downlink laser sub-signals into second downlink electrical signals and divide the second downlink electrical signals into a first signal **F4**, a second signal **F5**, and a third signal, and perform radio frequency signal magnitude conditioning and radio frequency amplification on the third signal in sequence and transmit the third signal. The multi-standard remote radio units **3** are further configured to filter a received second wireless signal or switch the radio frequency switch to obtain an uplink radio frequency signal, perform low-noise amplification and radio frequency signal magnitude conditioning on the uplink radio frequency signal in sequence and combine the uplink radio frequency signal with a second radio frequency signal **F6** and convert the same into an uplink laser signal, and transmit the uplink laser signal to the second optical fiber interfaces **40**.

[0060] Referring to FIG. **2**, the signal access intelligent unit **1** further includes a first radio frequency switch/filter **8**, a first downlink ATT **9**, a downlink amplifier **10**, a first signal coupler **11**, a large dynamic ROF optical transmitter **12**, an OOK modulator **13**, a wavelength division multiplexer **14**, a first power supply module **15**, a first communication module **16**, a master monitor unit **17**, a first uplink ATT **18**, a first uplink amplifier **19**, a second signal coupler **20**, and a first ROF optical receiver **21**.

[0061] Input and output ports of the first radio frequency switch/filter **8** are electrically connected to a second port of the radio frequency interface **38**. For example, the input and output ports of the first radio frequency switch/filter **8** are electrically connected to the second port of the radio frequency interface **38** through a first radio frequency cable.

[0062] An input port of the first downlink ATT **9** is electrically connected to the output port of the first radio frequency switch/filter **8**. For example, the input port of the first downlink ATT **9** is electrically connected to the output port of the first radio frequency switch/filter **8** through a second radio frequency cable.

[0063] An input port of the downlink amplifier **10** is electrically connected to an output port of the

first downlink ATT **9**. For example, the input port of the downlink amplifier **10** is electrically connected to the output port of the first downlink ATT **9** through a third radio frequency cable.

[0064] A first input port of the first signal coupler **11** is electrically connected to an output port of the downlink amplifier **10**.

[0065] An input port of the large dynamic ROF optical transmitter **12** is electrically connected to an output port of the first signal coupler **11**.

[0066] A radio frequency output port of the OOK modulator **13** is electrically connected to an input port of the first signal coupler **11**.

[0067] An input port of the wavelength division multiplexer **14** is connected to an output port of the large dynamic ROF optical transmitter **12** through a fourth optical fiber and input and output ports of the wavelength division multiplexer **14** are connected to the first optical fiber interface **39** through a fifth optical fiber.

[0068] An input port of the first ROF optical receiver **21** is connected to an output port of the wavelength division multiplexer **14** through a sixth optical fiber.

[0069] An input port of the second signal coupler **20** is electrically connected to an output port of the first ROF optical receiver **21**.

[0070] An Input port of the first uplink amplifier **19** is electrically connected to a first output port of the second signal coupler **20**.

[0071] An input port of the first uplink ATT **18** is electrically connected to an output port of the first uplink amplifier **19**. For example, the input port of the first uplink ATT **18** is electrically connected to the output port of the first uplink amplifier **19** through a fourth radio frequency cable. An output port of the first uplink ATT **18** is electrically connected to the input port of the first radio frequency switch/filter **8**. For example, the output port of the first uplink ATT **18** is electrically connected to the input port of the first radio frequency switch/filter **8** through a fifth radio frequency cable.

[0072] An input port of the first communication module **16** is electrically connected to a second output port of the second signal coupler **20** and an output port of the first communication module **16** is electrically connected to a third input port of the first signal coupler **11**.

[0073] A data port of the master monitor unit **17** is connected to a data port of the first radio frequency switch/filter **8**, a data port of the first downlink ATT **9**, a data port of the downlink amplifier **10**, a data port of the large dynamic ROF optical transmitter **12**, a data port of the OOK modulator **13**, a data port of the first communication module **16**, a data port of the first uplink ATT **18**, a data port of the first uplink amplifier **19**, and a data port of the first ROF optical receiver **21** through a first data line.

[0074] Referring to FIG. **1**, the optical fiber splitter **2** further includes a $1\times N$ optical splitter **4** and a 1×8 optical splitter **5**.

[0075] An input optical port of the $1\times N$ optical splitter **4** serves as the first optical port **42** of the optical fiber splitter **2**.

[0076] There are a plurality of 1×8 optical splitters **5**, where input optical ports of the 1×8 optical splitters **5** are connected to output optical ports of the $1\times N$ optical splitter **4** in a one-to-one correspondence manner through the third optical fiber and output optical ports of the 1×8 optical splitters **5** serve as the second optical ports **43** of the optical fiber splitter **2**.

[0077] Referring to FIG. **3**, the multi-standard remote radio unit **3** further includes a wavelength division demultiplexer **22**, a second ROF optical receiver **23**, a third signal coupler **24**, a second downlink ATT **25**, a downlink PA **26**, an OOK demodulator **27**, a slave monitor unit **28**, a second communication module **29**, a second power supply module **30**, a second radio frequency switch/filter **31**, a filter **32**, an antenna **33**, an uplink LNA **34**, a second uplink ATT **35**, a fourth signal coupler **36**, and a variable wavelength ROF optical transmitter **37**.

[0078] Input and output ports of the wavelength division demultiplexer **22** are connected to the second optical fiber interface **40** through a tenth optical fiber.

[0079] An input port of the second ROF optical receiver **23** is electrically connected to an output port of the wavelength division demultiplexer **22** through an eighth optical fiber.

[0080] An input port of the third signal coupler **24** is electrically connected to an output port of the second ROF optical receiver **23**. For example, the input port of the third signal coupler **24** is electrically connected to the output port of the second ROF optical receiver **23** through a sixth radio frequency cable.

[0081] An input port of the second downlink ATT **25** is electrically connected to a first output port of the third signal coupler **24**.

[0082] An input port of the downlink PA **26** is electrically connected to an output port of the second downlink ATT **25**. For example, the input port of the downlink PA **26** is electrically connected to the output port of the second downlink ATT **25** through a seventh radio frequency cable.

[0083] An input port of the second radio frequency switch/filter **31** is electrically connected to an output port of the downlink PA **26**. For example, the input port of the second radio frequency switch/filter **31** is electrically connected to the output port of the downlink PA **26** through an eighth radio frequency cable.

[0084] A first port of the filter **32** is electrically connected to output and output ports of the second radio frequency switch/filter **31**. For example, the first port of the filter **32** is electrically connected to the input and output ports of the second radio frequency switch/filter **31** through a ninth radio frequency cable.

[0085] The antenna **33** is electrically connected to a second port of the filter **32**.

[0086] An input port of the uplink LNA **34** is electrically connected to the output port of the second radio frequency switch/filter **31**. For example, the first port of the filter **32** is electrically connected to the input and output ports of the second radio frequency switch/filter **31** through a tenth radio frequency cable.

[0087] An input port of the second uplink ATT **35** is electrically connected to an output port of the uplink LNA **34**. For example, the first port of the filter **32** is electrically connected to the input and output ports of the second radio frequency switch/filter **31** through an eleventh radio frequency cable.

[0088] A first input port of the fourth signal coupler **36** is electrically connected to an output port of the second uplink ATT **35**.

[0089] An input port of the variable wavelength ROF optical transmitter **37** is electrically connected to an output port of the fourth signal coupler **36** and an output port of the variable wavelength ROF optical transmitter **37** is electrically connected to the input port of the wavelength division demultiplexer **22** through a ninth optical fiber.

[0090] An input port of the OOK demodulator **27** is electrically connected to a second output port of the third signal coupler **24**.

[0091] An input port of the second communication module **29** is electrically connected to a third output port of the third signal coupler **24**, and an output port of the second communication module **29** is electrically connected to the second input port of the fourth signal coupler **36**.

[0092] A data port of the slave monitor unit **28** is connected to a data port of the second ROF optical receiver **23**, a data port of the second downlink ATT **25**, a data port of the downlink PA **26**, a data port of the OOK demodulator **27**, a data port of the second communication module **29**, a data port of the uplink LNA **34**, a data port of the second uplink ATT **35**, and a data port of the variable wavelength ROF optical transmitter **37** through a second data line.

[0093] As shown in FIG. 2, FIG. 2 is a functional block diagram of the signal access intelligent unit **1**. The first wireless signal such as 4G/5G/WIFI is inputted into the first radio frequency switch/filter **8** through the radio frequency interface **38**, i.e., the first wireless signal such as 4G/5G/WIFI is inputted from the first port of the radio frequency interface **38**, enters the input and output ports of the first radio frequency switch/filter **8** after being outputted from the second port of the radio frequency interface **38**, and is transmitted and filtered by the first radio frequency

switch/filter **8** to obtain a downlink signal, the downlink signal is transmitted through a second radio frequency line to the first downlink ATT **9** for radio frequency signal magnitude conditioning, and then enters the downlink amplifier **10** for radio frequency amplification to obtain a first downlink electrical signal, the first downlink electrical signal, the OOK modulating signal **F1** outputted by the OOK demodulator **13**, and the first radio frequency signal **F2** outputted by the first communication module **16** are combined together through the first signal coupler **11** to obtain a fourth radio frequency signal, the fourth radio frequency signal is transmitted to the large dynamic ROF optical transmitter **12**, the large dynamic ROF optical transmitter **12** converts the fourth radio frequency signal into a downlink laser signal, and the downlink laser signal and the resultant uplink laser signal are multiplexed in one optical fiber (the fifth optical fiber) through the wavelength division multiplexer **14**, and the signal is transmitted from a long distance to the multi-standard remote radio unit **3** through the first optical fiber, the optical fiber splitter **2**, and the second optical fiber in sequence from the optical fiber interface **39**. The resultant uplink laser signal transmitted from the multi-standard remote radio unit **3** is decomposed by the wavelength division multiplexer **14** and transmitted to the first ROF optical receiver **21**, the first ROF optical receiver **21** converts the resultant uplink laser signal into an uplink electrical signal, couples a signal **F3** through the second signal coupler **20** and transmits the signal to the first communication module **16**, and other uplink electrical signals are outputted to the first uplink amplifier **19** for radio frequency signal amplification, are then subjected to signal magnitude conditioning through the first uplink ATT **18**, and finally, are outputted by the first radio frequency switch/filter **8** to the radio frequency interface **38** and are externally transmitted to a 4G/5G/WIFI signal source. The master monitor unit **17** in the signal access intelligent unit **1** is a central processing unit, responsible for a control command of the user and a synchronizing signal of the 4G/5G/WIFI signal source. Settings of devices in the system are controlled according to a system algorithm, parameters thereof are monitored, monitoring parameters are saved, alarms of the device are collected, and data is uploaded timely to a user monitor center. The master monitor unit **17** is further configured to modulate monitor data of the device into a first radio frequency signal **F2** through the first communication module **16**, modulate a synchronizing signal of the 4G/5G/WIFI signal source to an OOK modulating signal **F1** through the OOK modulator **13**, then modulate the first radio frequency signal **F2**, the OOK modulating signal **F1**, and the first downlink electrical signal together to the downlink laser signal for optical fiber transmission, and monitor state information of the multi-standard remote radio unit **3**. The master monitor unit **17** is configured to set parameters of the multi-standard remote radio unit **3**. For a downlink, the master monitor unit **17** reads the downlink output optical power from the large dynamic ROF optical transmitter **12** and compares the optical power value with a preset target optical power value. When an error is greater than a first threshold, the first downlink ATT **9** is correspondingly adjusted. The master monitor unit **17** and a monitor data packet communicated with the slave monitor unit **28** are modulated to the first radio frequency signal **F2** by the first communication module **16** through the first data line and then are coupled to the large dynamic ROF optical transmitter **12** through the first signal coupler **11** to form the downlink laser signal for transmission. For an uplink, the resultant uplink laser signal is demodulated by the first communication module **16** to obtain monitor data sent by the slave monitor unit **28** and is transmitted to the master monitor unit **17** through the first data line, and the master monitor unit **17** interprets the monitor data to perform a corresponding operation. In addition, the master monitor unit **17** further needs to monitor the output power of the first uplink amplifier to determine whether the uplink output power is in a target range. If an error is greater than a second threshold, the first uplink ATT **18** is correspondingly adjusted.

[0094] The radio frequency interface **38** is configured to externally link a 4G/5G/WIFI signal source device. There are two options for the first radio frequency switch/filter **8**. One is a radio frequency switch, which switches the uplink signal and the downlink signal by way of switching and is used in a TDD communication system. The other one is to separate the uplink signal from

the downlink signal by using a filter and is used in an FDD communication system. The first downlink ATT **9** is configured to condition the magnitude of the first wireless signal. The downlink amplifier **10** is configured to amplify the first wireless signal subjected to signal magnitude conditioning by the first downlink ATT **9**, so as to obtain the first downlink electrical signal. The first signal coupler **11** is configured to couple the signal and couple the OOK modulating signal F1 outputted by the OOK modulator **13** and the first radio frequency signal F2 outputted by the first communication module **16** into the downlink. The large dynamic ROF optical transmitter **12** is configured to convert the fourth radio frequency signal into the downlink laser signal and externally transmit the signal through the fourth optical fiber. The wavelength division multiplexer **14** is configured to perform wavelength division multiplexing on the downlink laser signal and the resultant uplink laser signal of the large dynamic ROF optical transmitter **12** into one optical fiber, i.e., the fifth optical fiber, and output the signal to the first optical fiber interface **39**. The first optical fiber interface **39** is an interface connected to an external optical fiber network. The first ROF optical receiver **21** is configured to receive the resultant uplink laser signal, convert the resultant uplink laser signal into the uplink electrical signal, and output the uplink electrical signal to the second signal coupler **20**. The second signal coupler **20** is configured to couple a small part of signal F3 to the first communication module **16**. The first uplink amplifier **19** is configured to amplify the uplink electrical signal. The first uplink ATT **18** is configured to condition the magnitude of the uplink electrical signal amplified by the first uplink amplifier **19**. The master monitor unit **17** is a processor of the signal access intelligent unit **1**, is configured to control the modules, set and monitor the modules, and save monitor information. The master monitor unit **17** is configured to transmit the monitor data and modulate the monitor data signal to the first radio frequency signal F2 through the first communication module **16** and externally transmit and receive the signal. The OOK modulator **13** is configured to modulate the synchronizing signal of the TDD wireless signal to the OOK modulating signal F1 and externally transmit the signal into the first signal coupler **11**. The first power supply module **15** is configured to supply power to active devices.

[0095] As shown in FIG. **1**, the optical fiber splitter **2** may be a common optical fiber splitter with one optical fiber at present and has no requirements on an optical process as long as conventional technical indexes on the current market may be realized. Thus, the cost is very low, it is quite convenient to wire and construct, and a current fiber-to-the-home PON network may also be used for low cost signal coverage.

[0096] The optical fiber splitter **2** includes two-stage optical splitters, a first optical port **42**, and a plurality of second ports **43**. The first stage is the $1 \times N$ optical splitter **4**, and the second stage is the 1×8 optical splitter **5**. If the first stage $1 \times N$ optical splitter **4** is replaced with an optical fiber patch cord, the second stage uses one 1×8 optical splitter **5**, thereby forming a one-in-eight network form; if the first stage $1 \times N$ optical splitter **4** uses a 1×2 optical splitter, the second stage will use two 1×8 optical splitters **5**, thereby forming a one-in-sixteen network form; if the first stage $1 \times N$ optical splitter **4** uses a 1×3 optical splitter, the second stage will use three 1×8 optical splitters **5**, thereby forming a one-in-twenty four network form; and by parity of reasoning, a one-in-sixty four network form may be formed at most. These optical splitters may be centralized in one place or may be separately placed in different places to facilitate construction. According to the present disclosure, one signal access intelligent unit **1** performs wireless signal coverage with 64 multi-standard remote radio units **3** through one optical fiber.

[0097] Since the optical fiber splitter **2** is reversible, besides outputting the laser, it may also input the laser. At least one uplink laser signal is inputted to the 1×8 optical splitter **5** through at least one second optical port **43** by the second optical fibers in a one-to-one correspondence manner, and together with the rest 7 uplink laser signals, are combined and transmitted to the $1 \times N$ optical splitter **4** through the third optical fiber, and then the $1 \times N$ optical splitter **4** combines the laser signals outputted by N 1×8 optical splitters **5** into one resultant uplink laser signal and outputs the

resultant uplink laser signal to the first optical port **42**, and transmits the resultant uplink laser signal to the signal access intelligent unit **1** through the first optical fiber. N is an integer greater than or equal to 1, and the specific number is determined according to an actual demand.

[0098] The $1 \times N$ optical splitter **4** averagely divides one beam of light into N parts by way of optically splitting one optical fiber. The first stage optical splitter is at most 8 parts, the power attenuation is 9 dB. In addition to the additional loss, the power attenuation is budged by 10 dB.

[0099] The 1×8 optical splitter **5** averagely divides one beam of light into 8 parts by way of optically splitting one optical fiber. The power attenuation is 9 dB. In addition to the additional loss, the power attenuation is budged by 10 dB.

[0100] As shown in FIG. **3**, the multi-standard remote radio unit **3** decomposes the downlink laser signal transmitted by the optical fiber splitter **2** through the wavelength division demultiplexer **22** and outputs the downlink laser signal to the second ROF optical receiver **23**. The second ROF optical receiver **23** converts the downlink laser signal into the second downlink electrical signal and outputs the second downlink electrical signal to the third signal coupler **24**. The third signal coupler **24** divides the second downlink electrical signal into a first signal, a second signal, and a third signal. The first signal is the signal F**4**, which is transmitted to the OOK demodulator **27** to demodulate the synchronizing signal of the wireless signal; the second signal is the signal F**5**, which is transmitted to the second communication module **29**, and the monitor information of the signal access intelligent unit **1** is demodulated and transmitted to the slave monitor unit **28**; the third signal is the second downlink electrical signal coupled by the third signal coupler **24**, is outputted to the second downlink ATT **25** for radio frequency signal magnitude conditioning, is then subjected to radio frequency signal magnitude conditioning through the downlink PA **26**, is then outputted to the second radio frequency switch/filter **31**, is then subjected to radio frequency filtering through the filter **32**, and finally, is outputted to the antenna **33**. The antenna transmits the filtered second downlink electrical signal to perform wireless signal coverage such as 4G/5G/WIFI for the covering space. The uplink radio frequency signal is extracted from the second wireless signal such as 4G/5G/WIFI signal of the user received by the antenna **33** through the filter **32** and the second radio frequency switch/filter **31**. The uplink radio frequency signal is subjected to low noise amplification through the uplink LNA **34** and is then subjected to radio frequency signal conditioning through the second uplink ATT **35**, the uplink radio frequency signal subjected to radio frequency signal conditioning by the second uplink ATT **35** and the second radio frequency signal F**6** outputted by the second communication module **29** are coupled together through the fourth signal coupler **36** to obtain the third radio frequency signal. The third radio frequency signal is outputted to the variable wavelength ROF optical transmitter **37** for electro-optical conversion to convert the third radio frequency signal into the uplink laser signal. The uplink laser signal is multiplexed in one optical fiber, i.e., the tenth optical fiber through the wavelength division demultiplexer **22** for transmission. The slave monitor unit **28** of the multi-standard remote radio unit **3** controls parameters of the assemblies, monitors monitor information of the assemblies, and saves them in a memory. Moreover, the slave monitor unit transmits the monitored information to the signal access intelligent unit **1** through the second communication module **29** in real time. The second power supply module **30** of the multi-standard remote radio unit **3** supplies power to the active devices of the device.

[0101] The second optical fiber interface **40** is configured to be connected to an optical interface of an external optical fiber network. The wavelength division demultiplexer **22** is configured to decompose the downlink laser signal and multiplex the resultant uplink laser signal. The second ROF optical receiver **23** is configured to receive the downlink laser signal, convert the downlink laser signal into the second downlink electrical signal, and output the second downlink electrical signal to the third signal coupler **24**. The third signal coupler **24** is configured to couple a small part of signals (signals F**4** and F**5**) to the second communication module **29** and the OOK demodulator **27**, respectively. The second downlink ATT **25** is configured to condition the magnitude of the third

signal. The downlink PA **26** is configured to amplify the third signal. The OOK demodulator **27** is configured to demodulate the OOK signal and demodulate the synchronizing signal of the first wireless signal. The slave monitor unit **28** is a processor of the multi-standard remote radio unit **3**, is configured to control the modules, set and monitor the modules, and save monitor information. The slave monitor unit may further transmit the monitor data. The second communication module **29** is configured to modulate the data signal of the slave monitor unit **28** to the second radio frequency signal F6 and externally transmit and receive the signal. In the downlink, the slave monitor unit **28** monitors the output power of the downlink PA **26** and determines whether the output power is in a target range. If an error is greater than a third threshold, the second uplink ATT **25** is adjusted. Moreover, the slave monitor unit **28** receives the monitor data demodulated by the second communication module **29** and sent by the master control unit **17**, interprets the monitor data, and performs a corresponding operation. If receiving a wavelength distribution command sent by the master monitor unit **17**, the slave monitor unit **28** will set the variable wavelength ROF optical transmitter **37** through the second data line, so that the variable wavelength ROF optical transmitter works at an appointed optical wavelength. In the uplink, the slave monitor unit **28** monitors the output power of the uplink LNA **34** and determines whether the output power is in a target range. If an error is greater than a fourth threshold, the slave monitor unit **28** adjusts the second uplink ATT **35** through the second data line. The second power supply module **30** is configured to supply power to active devices. There are two options for the second radio frequency switch/filter **31**. One is a radio frequency switch, which switches the uplink signal and the downlink signal by way of switching and is used in a TDD communication system. The other one is to separate the uplink signal from the downlink signal by using a filter and is used in an FDD communication system. The filter **32** filters the second wireless signal and the second downlink electrical signal to filter out out-of-band spurious signals. The antenna **33** is configured to receive the second wireless signal and transmit the second downlink electrical signal. The uplink LNA **34** is configured to perform low noise amplification on the uplink radio frequency signal. The second uplink ATT **35** is configured to condition the magnitude of the uplink radio frequency signal. The fourth signal coupler **36** is configured to combine the second radio frequency signal F6 outputted by the second communication module **29** and the uplink radio frequency signal together to obtain the third radio frequency signal. The variable wavelength ROF optical transmitter **37** is configured to convert the third radio frequency signal into the uplink laser signal and further has a wavelength tuning function.

[0102] An implementation of system wavelength distribution management will be described in detail below.

[0103] In the implementation of the wavelength distribution management, as shown in FIG. 1, FIG. 2, and FIG. 3, the signal access intelligent unit **1** and the multi-standard remote radio unit **3** are respectively provided with monitor units, where the monitor unit in the signal access intelligent unit **1** is the master monitor unit **17** and the monitor unit in the multi-standard remote radio unit **3** is the slave monitor unit **28**. The master monitor unit **17** is responsible for managing and distributing the optical wavelength, and the slave monitor unit **28** matches with the master monitor unit **17** to deploy the optical wavelength.

[0104] A specific distribution process of the optical wavelength is as follows:

[0105] In an initial state, the master monitor unit **17** maintains a wavelength resource state, as shown in table 1.

TABLE-US-00001 TABLE 1 Wavelength resource state table Wavelength Using state λ .sub.0 System reserved as a request channel λ .sub.1 Used λ .sub.n-1 Unused

[0106] In the above table, **20** is used as an initial wavelength request of the system, n is an integer greater than or equal to 1, and is the greatest **64**.

[0107] In the initial state, λ .sub.1, λ .sub.2, . . . , λ .sub.n-1 in the wavelength resource state maintained by the master monitor unit **17** all are identified as “unused” states. When the master

monitor unit **17** makes a response to the wavelength distribution request and distributes the corresponding wavelength to the slave monitor unit, in this case, the master monitor unit **17** will identify the using state of the wavelength as “used”. After the wavelength resource is recycled, it will be identified as “unused”.

[0108] In the operating process, the master monitor unit **17** and the slave monitor unit **28** detect the working state of the channel. When the master monitor unit **17** detects that a certain wavelength channel works abnormally, the wavelength resource is recycled and the using state thereof is identified as “unused”. When detecting that the current wavelength channel works abnormally, the slave monitor unit **28** will initiate a wavelength distribution request flow again.

[0109] The slave monitor unit **28** in the multi-standard remote radio unit **3** powered on is in the initial state, and in this case, the working wavelength of the multi-standard remote radio unit **3** is not distributed. The multi-standard remote radio unit communicates with the master monitor unit by using the system reserved wavelength **20** to trigger the wavelength distribution request flow. A specific process is as follows: [0110] 1) the slave monitor unit **28** initiates a wavelength distribution request message at a wavelength **20** channel and starts a message response timeout timer; [0111] 2) the master monitor unit **17** receives the wavelength distribution request message and verify the integrity of the message; [0112] (1) if the message is integral, the master monitor unit **17** distributes an unused wavelength to the slave monitor unit **28** from the optical wavelength resource table according to a certain principle; [0113] (2) if the message is not integral, the master monitor unit **17** does not respond; [0114] 3) the slave monitor unit **28** receives a response message from the master monitor unit **17** within a fixed time of the message response timer, and if the message is integral, the slave monitor unit responds to the master monitor unit **17** with a successful message, and sets a newly distributed wavelength where the multi-standard remote radio unit **3** works in the response message, as shown in FIG. 4; [0115] 4) if the slave monitor unit **28** responds to the message timer in a timeout manner or the response message is not integral, in order to avoid conflicts with another message of the slave monitor unit **28**, the slave monitor unit **28** delays random time and re-starts the wavelength distribution request flow, as shown in FIG. 5.

[0116] Compared with the prior art, the present disclosure has the following advantages and beneficial effects:

[0117] The present disclosure uses the ROF technology rather than ADC and DAC conversion, digital baseband processing, a digital optical module, and a digital HUB (router), and simulates passthrough. The working bandwidth reaches up to 10 MHz to 18 GHz. 4G, 5G, WIFI and millimeter 5G signals may be optically transmitted together to the multi-standard remote radio unit **3** for radio frequency amplification respectively to perform multi-standard signal coverage in the coverage field. According to the advantage that the working bandwidth in the ROF technology is 10 MHz to 18 GHz, the highest instant bandwidth of the wireless signal is 800 MHz, and signal coverage of 2×2MIMO, 4×4MIMO, and millimeter 5G is realized by way of frequency shift in the system. The optical transmission optical fiber network in the present disclosure only uses a common optical splitting network with one optical fiber to realize at most 64 multi-standard remote radio units **3**, as shown in FIG. 1.

[0118] Referring to FIG. 6, the digital optical fiber proximal end machine in a conventional digital wireless signal coverage solution uses digital ADC and DAC, digital baseband processing, and the digital optical module; and the digital optical fiber remote terminal machine uses digital ADC and DAC, digital baseband processing, and the digital optical module. In order to expand the units, a digital HUB (exchanger) is used. In order to expand 32 units, 4 digital HUBs are needed at most. The whole system is complicated. To transmit 4G, 5G, WIFI, and millimeter 5G signals together, the digital transmission rate reaches up to 400 Gb/s. The material cost is very high. The system of the remote terminal is hardly miniaturized, and the power consumption is relatively high.

[0119] Through comprehensive comparison, the optical fiber distributed wireless signal coverage system based on an ROF technology is simpler than the conventional digital wireless signal

coverage solution in architecture without high-speed ADC and DAC, digital baseband processing, high-speed digital optical module, high-speed HUB but transmits wireless signals such as 4G, 5G, WIFI, and millimeter 5G signals together only with the large dynamic ROF optical emission device, the ROF optical receiver, the variable wavelength ROF optical transmitter, other conventional radio frequency amplifiers, switches, filters, and the like. The system is convenient to design and construct, low in construction input, low in comprehensive cost, and suitable for large-area wireless signal coverage.

[0120] By using the large dynamic ROF optical transmission technology in the present disclosure, the input P-1 of the optical link reaches up to 32 dBm, the input IP3 thereof reaches up to 50 dBm, the transmitted spurious free dynamic range (SFDR) reaches up to 123 dB/Hz.^{sup.2/3}, the transmitted optical power reaches up to 50 mW, about 17 dBm, the optical insertion (20 dB optical insertion loss caused by 64 split paths of one optical fiber and 6 dB optical insertion loss caused by a 20 km optical fiber transmitted at 1550 nm, totally 26 dB optical insertion loss) of 26 dB in the optical link is still-9 dBm in the ROF optical receiver **23**, the signal is then subjected to radio frequency amplification by the subsequent amplifier, and the 5G signal meets the 3GPP requirement through actual measurement, which is the most apparent place in the technical advantages of the conventional solution.

[0121] FIG. **7** is a curve graph where third-order intercept point data is actually measured through a direct connection between a large dynamic ROF optical transmitter and an ROF optical receiver with 1 m optical fiber in the present disclosure; FIG. **8** is a curve graph where gain data is actually measured through the direct connection between a large dynamic ROF optical transmitter and an ROF optical receiver with 1 m optical fiber in the present disclosure; FIG. **9** is a curve graph where noise figure data is actually measured through the direct connection between a large dynamic ROF optical transmitter and an ROF optical receiver with 1 m optical fiber in the present disclosure; and FIG. **10** is a curve graph where spurious-free dynamic range data is actually measured through a direct connection between a large dynamic ROF optical transmitter and an ROF optical receiver with 1 m optical fiber in the present disclosure.

[0122] The highest radio frequency bandwidth of the large dynamic ROF optical transmission link may reach 18 GHz.

[0123] The variable wavelength ROF optical transmission technology is used in the present disclosure. Since device terminals of at most 64 multi-standard remote radio units **3** are accessed by the optical splitting network with one optical fiber, if the 64 multi-standard remote radio units **3** use the uplink laser with the same wavelength, 64 uplink lasers form optical interference in the optical splitting network, so that the entire system cannot work normally. The signal access intelligent unit **1** in the present disclosure automatically distributes the wavelength of one uplink laser for each of the multi-standard remote radio units **3** according to the number of the multi-standard remote radio units **3**, ensuring the wavelength uniqueness of each of the multi-standard remote radio units **3**. Different from the wavelengths of the other multi-standard remote radio units **3**, the optical interference of the 64 uplink lasers in the optical splitting network is avoided, which is the core technology of the present disclosure.

Claims

1. An optical fiber distributed wireless signal coverage system based on a radio over fiber (ROF) technology, comprising: a signal access intelligent unit (**1**), comprising a radio frequency interface (**38**) and a first optical fiber interface (**39**) and configured to perform radio frequency signal magnitude conditioning and then radio frequency amplification on a first wireless signal received from the radio frequency interface (**38**) to obtain a first downlink electrical signal and combine the first downlink electrical signal, an OOK modulating signal (F1), and a first radio frequency signal (F2) together and convert the same into a downlink laser signal, and output the downlink laser

signal to the first optical fiber interface (39); and further configured to convert a resultant uplink laser signal inputted from the first optical fiber interface (39) into an uplink electrical signal and perform radio frequency amplification and radio frequency signal magnitude conditioning on the uplink electrical signal in sequence to output the uplink electrical signal from the radio frequency interface (38); an optical fiber splitter (2), comprising a first optical port (42) and at least one second optical port (43), wherein the first optical port (42) thereof is connected to the first optical fiber interface (39) through a first optical fiber; configured to averagely divide the downlink laser signal inputted through the first optical port (42) into a plurality of downlink laser sub-signals, wherein the downlink laser sub-signals correspond to the second optical ports (43) one by one and are outputted from the second optical ports (43); and further configured to combine the uplink laser signals inputted through the second optical ports (43) into a resultant uplink laser signal and output the resultant uplink laser signal from the first optical port (42); and at least one multi-standard remote radio unit (3), the multi-standard remote radio units comprising second optical fiber interfaces (40) respectively, wherein the second optical fiber interfaces (40) are respectively connected to the second optical ports (43) of the optical fiber splitter (2) in a one-to-one correspondence manner through a second optical fiber; configured to decompose the downlink laser sub-signals and convert the downlink laser sub-signals into second downlink electrical signals and divide the second downlink electrical signals into a first signal (F4), a second signal (F5), and a third signal, and perform radio frequency signal magnitude conditioning and radio frequency amplification on the third signal in sequence and transmit the third signal; and further configured to filter a received second wireless signal to obtain an uplink radio frequency signal, perform low-noise amplification and radio frequency signal magnitude conditioning on the uplink radio frequency signal in sequence and combine the uplink radio frequency signal with a second radio frequency signal and convert the same into an uplink laser signal, and transmit the uplink laser signal to the second optical fiber interfaces (40).

2. The optical fiber distributed wireless signal coverage system based on a ROF technology according to claim 1, wherein the signal access intelligent unit (1) further comprises: a first radio frequency switch/filter (8), input and output ports thereof being electrically connected to a second port of the radio frequency interface (38); a first downlink ATT (9), an input port thereof being electrically connected to the output port of the first radio frequency switch/filter (8); a downlink amplifier (10), an input port thereof being electrically connected to an output port of the first downlink ATT (9); a first signal coupler (11), a first input port thereof being electrically connected to an output port of the downlink amplifier (10); a large dynamic ROF optical transmitter (12), an input port thereof being electrically connected to an output port of the first signal coupler (11); an OOK modulator (13), an output port thereof being electrically connected to a second input port of the first signal coupler (11); a wavelength division multiplexer (14), an input port thereof being connected to an output port of the large dynamic ROF optical transmitter (12) through a fourth optical fiber and input and output ports thereof being connected to the first optical fiber interface (39) through a fifth optical fiber; a first ROF optical receiver (21), an input port thereof being connected to an output port of the wavelength division multiplexer (14) through a sixth optical fiber; a second signal coupler (20), an input port thereof being electrically connected to an output port of the first ROF optical receiver (21); a first uplink amplifier (19), an input port thereof being electrically connected to a first output port of the second signal coupler (20); a first uplink ATT (18), an input port thereof being electrically connected to an output port of the first uplink amplifier (19) and an output port thereof being electrically connected to the input port of the first radio frequency switch/filter (8); a first communication module (16), an input port thereof being electrically connected to a second output port of the second signal coupler (20) and an output port thereof being electrically connected to a third input port of the first signal coupler (11); and a master monitor unit (17), a data port thereof being connected to a data port of the first radio frequency switch/filter (8), a data port of the first downlink ATT (9), a data port of the downlink

amplifier (10), a data port of the large dynamic ROF optical transmitter (12), a data port of the OOK modulator (13), a data port of the first communication module (16), a data port of the first uplink ATT (18), a data port of the first uplink amplifier (19), and a data port of the first ROF optical receiver (21) through a first data line.

3. The optical fiber distributed wireless signal coverage system based on a ROF technology according to claim 1, wherein the optical fiber splitter (2) further comprises: a $1 \times N$ optical splitter (4), an input optical port thereof serving as the first optical port (42) of the optical fiber splitter (2); and at least one 1×8 optical splitter (5), input optical ports thereof being connected to output optical ports of the $1 \times N$ optical splitter (4) in a one-to-one correspondence manner through the third optical fiber and output optical ports thereof serving as the second optical ports (43) of the optical fiber splitter (2).

4. The optical fiber distributed wireless signal coverage system based on a ROF technology according to claim 1, wherein the multi-standard remote radio unit (3) further comprises: a wavelength division demultiplexer (22), input and output ports thereof being connected to the second optical fiber interface (40) through a tenth optical fiber; a second ROF optical receiver (23), an input port thereof being connected to an output port of the wavelength division demultiplexer (22) through an eighth optical fiber; a third signal coupler (24), an input port thereof being electrically connected to an output port of the second ROF optical receiver (23); a second downlink ATT (25), an input port thereof being electrically connected to a first output port of the third signal coupler (24); a downlink PA (26), an input port thereof being electrically connected to an output port of the second downlink ATT (25); a second radio frequency switch/filter (31), an input port thereof being electrically connected to an output port of the downlink PA (26); a filter (32), a first port thereof being electrically connected to input and output ports of the second radio frequency switch/filter (31); an antenna (33), electrically connected to a second port of the filter (32); an uplink LNA (34), an input port thereof being electrically connected to the output port of the second radio frequency switch/filter (31); a second uplink ATT (35), an input port thereof being electrically connected to an output port of the uplink LNA (34); a fourth signal coupler (36), a first input port thereof being electrically connected to an output port of the second uplink ATT (35); a variable wavelength ROF optical transmitter (37), an input port thereof being electrically connected to an output port of the fourth signal coupler (36) and an output port thereof being electrically connected to the input port of the wavelength division demultiplexer (22) through a ninth optical fiber; an OOK demodulator (27), an input port thereof being electrically connected to a second output port of the third signal coupler (24) and a data interface thereof being connected to the second downlink ATT (25) and the downlink PA (26) through a third data line, respectively; a second communication module (29), an input port thereof being electrically connected to a third output port of the third signal coupler (24); and a slave monitor unit (28), a data port thereof being connected to a data port of the second ROF optical receiver (23), a data port of the second downlink ATT (25), a data port of the downlink PA (26), a data port of the OOK demodulator (27), a data port of the second communication module (29), a data port of the uplink LNA (34), a data port of the second uplink ATT (35), and a data port of the variable wavelength ROF optical transmitter (37) through a second data line.
